

Exchange Rate Return Co-movements and Volatility Spillover: The Case of Emerging Market Economies

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Abstract

The paper examines the return co-movements and volatility spillover among four major foreign exchange markets and four emerging foreign exchange markets namely India, China, South Africa and Brazilian foreign exchange market. In particular the co-movement and spillover among Euro, Japanese Yen, Australian Dollar and Swiss Franc with the above mentioned emerging foreign exchange markets for the period 1995-2015 is the main objective of the study. Based on daily data, the paper estimates a flexible MGARCH- Dynamic Conditional Correlation model and VAR-based spillover index. The empirical investigation suggest the presence of significant return co-movement and volatility spillover between the foreign exchange markets with Euro as the net transmitter of volatility to Indian Rupee and Chinese Yuan and Australian Dollar as the net transmitter of volatility to Brazilian Real and South African Rand.

JEL Classification: F31, F41, C32

Key Words: Exchange Rate, Volatility, Return Co-movement, MGARCH, DCC Model, VAR based Spillover Index.

1. INTRODUCTION

The appearance that immediately importunes in the mind with the word ‘volatile’ is that of unstable stock market, balance of payments crisis of the late 1990s or the unpredictable capital flows in the emerging market economies. But over the last decade the perception of volatility had widened and began to develop itself into an independent field of inquiry in the foreign exchange market as well. Conceptually volatility can be decomposed into a predictable component and an unpredictable component. In common dialect, it is a arduous task to make a distinction among volatility, uncertainty, risk, variability, fluctuation or oscillation but according to Knight’s (1921) volatility is allied to risk in that it provides a measure of the possible variation or movement in a particular economic variable or some function of that variable over some historical period. Mainly there are two key essence of volatility namely variability and uncertainty - variability in an economic variable may be anticipated while the residual which captures pure risk or uncertainty is unanticipated and constitutes a ‘shock’. The size and persistence of such shocks can pose a major challenge to the economic management as it can evolve into extreme volatility or a crisis.

Volatility in exchange rate can be explained in the presence of three possible factors - volatility in market fundamentals, changes in expectations due to new information and speculative “bandwagons” (Engel and Hakkio 1993). Volatility in market fundamentals such as the money supply, income and interest rates affects volatility in exchange rate because exchange rate is a function of these fundamentals. Changes in expectations about future market fundamentals or economic policies also affect exchange rate volatility as new information induces the market participants to alter their forecasts of future economic conditions and policies thereby leading to exchange rate volatility. Finally volatility in exchange rate can be caused by speculative bandwagons or speculative exchange rate movement unrelated to current or expected market fundamentals. If speculators buy dollars on the assumption that dollar will appreciate but it becomes apparent that market fundamentals will not sustain such an appreciation, results in selling of the dollars which could cause the dollar to depreciate. Such behavior on the part of the speculator can cause the value of the dollar to fluctuate thereby contributing to exchange rate volatility.

In particular exchange rate is defined as the relative price of currencies between two or more countries, its behavior impacts the competitiveness of exports, international investment portfolios, international reserves and currency value of debt payments and more precisely impacts the overall stability of the economy. So any misalignment in the exchange rate thus requires intervention in the exchange market through appropriate exchange rate policy along with monetary and fiscal policies to ensure stability and growth in the economy. For instance, during the BOP crisis of 1991 in India, exchange rate management became a pivotal policy measure. The exchange rate system in India underwent changes towards a unified exchange rate system through merging of a dual exchange rate and finally towards current account convertibility in the early 1990s. The exchange rate regime shifted from a par value system to a basket-peg and further to a managed float exchange rate system over the period of time. The regime has evolved over time in tune with the changes in the global payments system towards a generalised float of the currency and gradual opening up of the domestic economy as part of the broader strategy of macroeconomic reforms since 1991. In March 1992, with the introduction of The Liberalized Exchange Rate Management System (LERMS) the dual exchange rate system was established. But in March 1993, as per the recommendations of the Rangarajan Committee, the dual exchange rate system was replaced by unified exchange rate system and the final step towards current account convertibility was taken in August 1994. Another move towards reform of the exchange rate regime was the relaxation of controls over foreign exchange transactions.

Furthermore in 1995, the Sodhani Committee on Foreign Exchange Markets in India made several recommendations concerning participants, trading, risk management and selective market intervention by the Reserve Bank to patronize the development in foreign exchange market in an organized way. This was followed by wide-ranging reforms in the Indian foreign exchange market in early 1996. There was a gradual opening of capital account and greater integration of the Indian financial markets with the global network since 2003-04. However, distinct occasions of volatility have been effectively managed by timely intervening in the market along with monetary and administrative measures.

On the other hand during the later part of 1970s, the Chinese exchange rate regime has changed periodically - first it was centralized and still a fixed one, the exchange rate was dual from 1986 to 1993 where the official rate that was subject to sporadic adjustments coexisted with

the market-determined rate in the swap centres whereas the second rate was set at a relatively depreciated level as compared to the official rate. In 1994 the official rate was devaluated and unified with the exchange rate at the swap centres and a single, managed floating exchange regime was officially adopted. In practice, however, after short period of some appreciation of nominal exchange rate, the renminbi (RMB), also known as yuan, was de facto fixed to the US dollar since 1995. Central bank's intervention to maintain a stable exchange rate has been large scale with reserves rising to more than 30% of GDP by end 2004. In July 2005, the Chinese authorities announced a revaluation of the exchange rate against the dollar of 2.1% with the currency able to move daily in a narrow range of $\pm 0.3\%$. In addition the central rate for the RMB is determined by a basket of currencies the composition of which was not revealed.

In case of Brazil, the economy was in a stage of low and volatile growth in 1980s. But in the early 1990s, the authorities followed a pattern of economic development based on crawling peg exchange rate regime with nominal diary devaluations that resulted in a depreciated real exchange rate and a policy of high real interest rates. Such policy generated both high trade balance surplus and the attraction of capital flows but at the costs of a very high inflation. Nevertheless in July 1994, the Real Plan, characterized by a crawling exchange rate band, was launched in order to stabilize the domestic currency in nominal terms by implementing a tight monetary policy. Accordingly a new unit of account, the Unit of Real Value (URV), latter re-named as Real, pegged to US dollar was introduced which operated as the new currency. Under this new exchange rate regime, the monetary authority was committed to selling foreign exchange reserves once the price of the Real reached one US dollar but was not obliged to intervene when the Real was worth more than one dollar. The effects of the Real Plan on the real exchange rate, the external accounts and debt accumulation process finally led to an exchange rate crisis in early 1999. Despite this devaluation of the Real, the BCB has had no commitment to determine exchange rate, although it has operated occasionally in the foreign exchange markets in periods of greater capital flows volatility and more recently in accordance to its policy to accumulate exchange reserves. Therefore, Brazilian exchange rate regime is not a freely floating exchange regime but it is closer to a floating exchange regime than other emerging market economies.

Following the Bretton Woods Agreement 1944, the South African pound was fixed at an exchange rate equal to £1 and US\$ 4.03. Henceforth the appointment of the Decimal Coinage Commission in 1956, the decimal system was pioneered in South Africa and on February 14 1961, a new system of rands and cents was introduced. Notwithstanding the fall of Bretton Woods Agreement in 1971, South Africa devalued the rand by 12.28 percent and \$1 was equal to 71.34c. Meanwhile the realignment of 1971 compelled the British government to allow the pound to float downward against other currencies from June 1972 and the rand was linked to this float. After four months, however, the rand was pegged to the US dollar and it remained pegged till 1974. In June 1974 the South African authorities decided to delink the rand from the dollar and a policy of independent managed floating was introduced and in June 1975 this was replaced by a fix against the dollar although the rand was devalued by 17.9 per cent to protect the balance of payments. To sum up, it should be noted that in the period consequent to the breakdown of the Bretton Woods Agreement, the exchange rate of the rand has, in turn, been pegged to the dollar, the pound sterling, the dollar, a basket of currencies and again against the dollar. It is noteworthy here that during this phase the level at which the rand was pegged to the dollar has been changed six times. Benefits, if any, of these many realignments are questionable.

On February 7 1983, the South African Government announced the abolition of the financial rand exchange rate system. But this relaxation of exchange control over nonresidents could not be sustained as the refusal of the key international banks to renew credit lines for South Africa forced the authorities to announce the temporary closure of the foreign exchange market six months later. On September 1 1985, a standstill on South Africa's international debt repayments was announced while exchange controls over nonresidents were reintroduced. A major feature of exchange rate management in the period from September 1985 until the period running up to democratic elections in the 1990s was the overriding objective of maintaining a surplus on the current account of the balance of payments. Since democratic elections in 1994 normality has returned to South Africa's international relations but the exchange rate of the rand against the US dollar has remained in the long term downward trend that commenced in the early 1980s.

In summary as the emerging markets progressed towards floating exchange regime and got more associated with the world financial market, the chances of volatility in exchange rate

had increased manifold and managing periods of such volatility has emerged as a great challenge in view of the impossible trinity of independent monetary policy, open capital account and exchange rate management for nearly a decade. Any purchase or sale of foreign assets by the central bank in a country can be treated as policy intervention in the foreign exchange market and it has important implications for exchange rate. For example in the post-Asian crisis period, particularly after 2002-03, capital flows into India surged creating space for speculation on Indian Rupee. During this period, the Reserve Bank, in an attempt to curb the volatility, directly intervened in the foreign exchange market through purchases and sales of the US Dollars and sterilized its impact on the monetary base. Thus the central bank intervenes from time to time to contain volatility arising due to demand-supply mismatch in the domestic foreign exchange market and maintain orderly market conditions to restore equilibrium exchange value. In this context it is important to note here, following Jansen and De Haan (2005), volatility effect on exchange rates are large with policy statements of European Central Bank. In case of India, Goyal and Arora (2010) show that communications by Reserve Bank have important implications for exchange rate volatility. Therefore on this backdrop it calls for understanding the nature of volatility and anticipate and manage its consequences should be of considerable interest to policymakers as empirical investigation increasingly shows that weak policies and institutions in developing countries may magnify the negative impact of volatility and can lead to a permanent setback relative to richer countries.

This study investigates the exchange rate return co-movement and volatility spillover between four emerging foreign exchange markets of India, China, South Africa and Brazil and several other currencies namely the Euro, Japanese Yen, Australian Dollar and Swiss Franc. It is important to note here that the notion of correlation reveals the nature of interdependencies among assets whereas the knowledge of spillover helps in understanding the proliferation of shocks from one market to another and how an unanticipated movement in one currency can affect the other currencies apart from macroeconomic fundamentals. On account of the presence of such volatility linkages across exchange rates, the literature embarks on a decision in the field of portfolio diversification, policy making and risk management. Although a variety of techniques could be applied to assess the return co-movement and volatility spillover but the most plausible among them is the Dynamic Conditional Correlation model and Vector

Autoregression framework. A more detailed account of volatility co-movement and spillover in exchange rates is presented in terms of a rigorous econometric exercise that follows later on in this paper.

The structure of the paper is as follows: Section 2 outlines the literature relevant to the study whereas section 3 describes the methodology and contains the empirical evaluation. Section 4 concludes the discussion with summary of findings and policy implications.

2. REVIEW OF LITERATURE

A large number of literatures have primarily focused on the stock market volatility spillover with little emphasis on the foreign exchange market spillover. However, in the past decade there came the necessity to examine the volatility in foreign exchange market as it became more associated with the world market. There is substantial literature on developed economies concerning the occurrence of volatility in exchange rates and its spillover effects to other foreign exchange markets with the papers emphasizing on the causes and methods of testifying volatility in exchange rate by employing GARCH family models. However, there is a dearth of literature in the context of spillover in case of emerging economies as majority of the papers have mainly concentrated on the role of intervention policy of the central bank in controlling volatility. A brief review of the existing papers is presented below.

While there are many approaches to measure volatility but the most common among them is the generalized autoregressive conditional heteroskedasticity model. GARCH estimates of volatility are calculated using a time series of past exchange rate changes. The basic GARCH family models are frequently applied and quoted to describe the volatility in financial markets such as stock exchanges and foreign exchange markets. Mundaca (1991) showed that GARCH models perform better than the ARCH model (Johnston and Scott, 2000 and Chong et al., 2002) whereas Sandoval (2006) captured the important characteristics of daily exchange rate for seven Asian and emerging Latin American countries by applying ARMA, GARCH, EGARCH and GJR-GARCH models. Stancik (2007) states that more openness leads to lower volatility, effect of news varies across countries and only key changes in exchange rate regimes have significant effect on exchange rate volatility. Giannellis and Papadopoulos (2011) use monetary, real and financial variables to assess the relevant importance of each of the variables to exchange rate

volatility in the case of selected EMU members and candidate countries. Kocenda and Valachy (2006), using a Threshold GARCH model, opined that volatility is greater under floating than under fixed regime. In another paper Erdemlioglu, Laurent and Neely (2012) modelled the volatility in exchange rate by incorporating intraday periodicity, autocorrelation and discontinuities in prices (Jumps) using high frequency data. The results show the prevalence of jumps in the data and stresses on accounting for intraday periodicity in detecting jumps. Another finding is that macro news affects volatility but not as much as periodicity caused by market openings and closings. Djeutem and Kasa (2013) shows that revision of robust forecast are more volatile than revisions of non-robust forecasts in the context of the monetary model of exchange rates. Annachhatre (2013) argues that exchange rate volatility is caused due to deviation from fundamentals, excessive speculative activities, macro-economic shocks or other global and domestic news. A major finding of the study is that fluctuations of Rupee against other major currencies were frequent after 1991 after the opening up of the economy. On the other hand Alam and Rahaman (2012) explored that both AR and ARMA models best suits the in-sample data while according to out-of-sample, GARCH and TARCH model best suits with and without transaction costs respectively. Moreover EGARCH and TARCH models outperform all the other models as per in-sample and out-of-sample trading performances are concerned. Similarly Kamal, Haq, Ghani & Khan (2012) also exhibited that EGARCH model best explains the volatile behavior of the daily exchange rate (Narayan et al., 2009). Mckenzie & Mitchell (2002) states that GARCH (1,1) model is preferred in case of symmetric reactions to the improvement of market. These studies indicate that the variance of daily exchange rate changes is forecastable using GARCH models.

Regarding volatility spillover, studies that examined exchange rate volatility transmission were initiated by Engle et al. (1990) where the authors found supporting evidence for two hypotheses namely the ‘heat waves’ and the ‘meteor shower’. Heat waves refer to exchange rate volatility in one particular market having only country specific effect while the meteor shower refer to volatility being transmitted to other countries. Accordingly, enquiry in this field were carried out by Bollerslev (1990), Speight and McMillan (2001), Melvin and Melvin (2003), Black and McMillan (2004) and Calvet et al. (2006). The main feature of these studies is the application of GARCH family models to assess volatility dependencies across currencies. The

financial literature postulates many possible reasons for such volatility spillovers across markets. Ross (1989) notes that apart from the information contained in prices themselves, volatility is also an important source of information in the financial markets. The first channel of volatility spillover is because of the common factors or news that affects a set of financial variables simultaneously (Bollerslev et al. 1992). It states that if there are fundamental linkages between the markets, variation in the common factors is likely to cause volatility spillover across markets. The second channel operates through the information spillover caused by the cross market hedging (Ederington & Lee 1993). Moreover the contagion hypothesis notes that agents who observe a price decline in one market becomes more risk averse and reduces their position in the other markets and thereby creating an apparent spillover effect (Ebrahim 2000). Anderson et al. (1999) constructed a model free estimate of daily exchange rate volatility and correlation and reported a normality-inducing volatility transmission, high contemporaneous correlation across volatilities, high correlation between correlation and volatility, pronounced and highly persistent temporal variation in both volatilities and correlation and clear evidence of long memory dynamics in both volatilities and correlation. Song (2009) indicated significant volatility spillover between Shanghai and Shenzhen and correlation contagion. Besides this Hongkong and Singapore presents a mildly high correlation. Antonakakis (2012) suggested that the Euro is the net transmitter of volatility while the Pound is the net receiver of volatility. Moreover the cross-market volatility spillovers are bidirectional and the highest spillovers occur between European markets. Nikkinen et al. (2006), in a VAR framework, investigated the daily implied volatility spillovers and found that the correlation is highest between the Euro and the Franc. In addition, Euro is the dominant currency in volatility transmission as the Euro was found to granger cause the Pound and the Franc but not vice-versa. Likewise Diebold and Yilmaz (2009) in a VAR framework found striking evidence of divergent behaviour in the dynamics of return spillovers and volatility spillovers in the context of nineteen global equity markets from the early 1990s. Diebold and Yilmaz (2012) also reviewed the volatility spillovers across US stock, bond, foreign exchange and commodities markets in a generalized VAR construction. Kearney and Patton (2000) pointed out both direct and indirect volatility transmission within the EMS and German mark was relatively insulated from outside shocks and held a dominant position in terms of volatility transmission. It is important to note here that the

results obtained using weekly data were less significant than those of daily data because of temporal aggregation of the data from daily to weekly which is in line with the conjecture that markets are more likely to transmit volatility in active phases rather than in calm ones (Ghose and Kroner 1996, Andersen and Bollerslev 1998). The study by Ghosh (2012) displayed that volatility has actually spilled over from stock market, government securities market, forward market, derivative market and international crude prices to the Indian foreign exchange market. Apart from this the stock market volatility emerged as the most important factor influencing volatility spillover in the foreign exchange market (Mishra et al., 2007). Behera (2011) signifies that Non-Deliverable-Forward market (NDF) shocks and volatilities influence the onshore markets. Sahoo (2012) marked the volatility transmission from Brazilian Real, the Russian Ruble, the South Korean Won, the Singapore Dollar, the Japanese Yen, the Swiss Franc, the British Pound Sterling and the Euro to the exchange rate of the Indian Rupee. Chowdhury and Sarno (2004) ascertained statistically significant intra-regional volatility spillover among the Central European foreign exchange market using a multivariate stochastic volatility model. Mukherjee (2011) had theorized that return volatility of the Indian equity market exhibits a sudden sharp increase and the conditional correlation of the equity return with all the other markets has increased over time. Saha and Chakrabarti (2011) displayed volatility spillover but no asymmetric impact between stock to exchange rate market and vice versa. Lee (2010) suggested the presence of both regional spillover and the transmission of shocks from external stock and foreign exchange markets. Hong (2001) proved the existence of granger causality between two weekly nominal US Dollar exchange rates with respect to the Deutsche Mark and the Japanese Yen. Cappiello et al. (2006) demonstrated that equity returns show strong evidence of asymmetries in conditional volatility with little is found for bond returns. However both equities and bonds exhibit asymmetries in conditional correlations with equities responding stronger than bonds to bad news. Horng and Chen (2010) unfolded that exchange rate volatility negatively affects Thailand's stock market and Japanese stock return volatility affects the variation risks in the Thailand's stock market. Lastly, Fang et al. (2006) suggest that within the domestic cross markets, the volatility transmission is unidirectional from the stock market to the bond market. But in case of international cross-market analysis, there is a strong evidence of

volatility spillover among the international stock markets than between international stock and bond markets.

A wide range of unending literature is furnished above which gives an idea about the occurrence of volatility in the foreign exchange markets. Nevertheless the preceding section also provides a view of the volatility co-movement and spillover among different currencies and the process of modelling that spillover and analyzing the consequences. Accordingly the next section will present a detailed econometric analysis of the volatility co-movement and interpret the results.

3. THE DATA AND METHODOLOGY

3.1. The Data

The study focuses on the period 1995-2015 using daily exchange rate of Euro, Japanese Yen, Australian Dollar, Swiss Franc, Indian Rupee, Chinese Yuan, South African Rand and Brazilian Real respectively. The rationale behind choosing Euro, Japanese Yen, Australian Dollar and Swiss Franc is that according to the Bank for International Settlements (BIS) (2013), these currencies are rated as the most traded currency among others. The BIS's Triennial Central Bank Survey on the turnover of these currencies in 2013 are \$ 1,785,720 million, \$ 1,231,249 million, \$ 4,61,689 million and \$ 2,75,472 million of all transactions including the spot transactions, outright forwards, foreign exchange swaps, currency swaps and foreign exchange options. It is also worth mentioning here that according to the European Central Bank, the conversion of European Currency Unit (ECU) to Euro on 1st January, 1999 was at 1:1 basis.

Before proceeding with the estimation, the variables were tested for non-stationary behaviour. The stationarity test based on Augmented Dickey-Fuller rejects the null hypothesis of unit roots in exchange rate at the first difference. The results of the ARIMA test confirms that all the variables follows AR(1) process. Further a detailed view of the descriptive statistics and the time path of the variables (Figure 1, Appendix) will help in signifying that most of them show abnormal movement during the period of analysis. The return of each foreign exchange is calculated by taking the first logarithmic differences in exchange rate denoted as:

$$\Delta \ln S_t = \ln S_t - \ln S_{t-1}$$

A close view at the return series of the variables reveals the presence of volatility clustering and ARCH effect which is supportive of modelling the volatility in exchange rate in a GARCH framework.

3.2 The Methodology:

Time-varying volatility models have been popular since the early 1990s in empirical research in finance. The analysis of volatility in financial market has been widely studied in ARCH-GARCH framework pioneered by Engle (1982) and further developed by Bollerslev (1986), Nelson (1991) and others. The most widely used models are the VECH model, the BEKK model and the Dynamic Conditional Correlation model. These models differ in their assumptions and specifications of the variance-covariance matrix, but helps in modelling time varying variance and covariance estimates.

In order to establish the presence of volatility in a financial time series, Autoregressive Conditional Heteroskedasticity (ARCH) and Generalised Autoregressive Conditional Heteroskedasticity (GARCH) methods are primarily used. The ARCH model, first presented by Engle (1982), shows that variance of the residuals at a particular time period depends on the squared error terms of the preceding period. The simplest ARCH (1,1) can be represented as

$$\sigma_t^2 = \gamma_0 + \gamma_1 u_{t-1}^2 \quad \dots(1)$$

The model can be expressed in a generalized form as ARCH (q) as

$$\sigma_t^2 = \gamma_0 + \sum_{j=1}^q \gamma_j u_{t-j}^2 \quad \dots(2)$$

The GARCH model, on the other hand, was first developed by Bollerslev (1986) as a step to check the drawback of ARCH model which, according to Engle, looked more like a moving average specification than an autoregression. The GARCH model illustrates that the variance scaling parameter not only depends on the past values of the error terms but also on the past values of itself.

The simplest GARCH (1, 1) can be represented as

$$\sigma_t^2 = \gamma_0 + \gamma_1 u_{t-1}^2 + \delta_1 \sigma_{t-1}^2 \quad \dots(3)$$

The above model can be expressed in a generalized form vis-à-vis GARCH (p, q)

$$\sigma_t^2 = \gamma_0 + \sum_{j=1}^q \gamma_j u_{t-j}^2 + \sum_{i=1}^p \delta_i \sigma_{t-i}^2 \quad \dots(4)$$

where p, q are the orders of the lagged squared error terms and lagged conditional variances respectively. If the value of $p = 0$ then the model reduces to ARCH (q). The parameter $\sum_{j=1}^q \gamma_j = \gamma$ is known as ARCH parameter and the parameter $\sum_{i=1}^p \delta_i = \delta$ is known as GARCH parameter. If the $\sum(\gamma + \delta)$ is close to one then it indicates the tendency of volatility to persist for longer time whereas if the summation is greater than one then it indicates the existence of explosive series with the tendency to move away from the mean value (Kuruville et al. 2012, Sekhar 2003). However, to investigate the volatility and co-movement, Multivariate GARCH models (Bauwens et al. 2006) was put in place as this method explicitly captures the time-varying covariance between two markets. In this paper the MGARCH-DCC model is estimated to take into account the time-varying nature and interrelations among the markets.

The Dynamic Conditional Correlation (DCC) model was proposed by Engle (2002) as an attempt to establish that correlations, both conditional and unconditional, among the markets are not constant rather time-varying. The model uses a two step procedure. In the first step, the individual conditional variances are specified as univariate GARCH processes and in the second step the correlation among the series is presented. This model have a computational advantage over other MGARCH models in that the number of parameters to be estimated in the process is independent of the number of series to be correlated. As a result, very large correlation matrices can be estimated. Nonetheless it is an investigation against the too restrictive assumption of constant correlation of Constant Conditional Correlation (CCC) Model and is represented as:

$$y_t = \mu_t(\theta) + \varepsilon_t \quad \dots(5)$$

$$H_t = D_t R_t D_t \quad \dots(6)$$

where $y_t = (y_{it} \dots y_{nt})'$ is a $n * 1$ vector of exchange rate return, $\mu_t(\theta) = (\mu_{it}, \dots, \mu_{nt})'$ is the conditional $n * 1$ mean vector of y_t , which can be specified as an ARMA process

$$\psi_i(L)(y_{it} - \mu_{it}) = \Theta_i(L)\varepsilon_t \quad \dots(7)$$

$$\mu_{it} = \mu_i + \sum_{j=1}^{n_1} \delta_{ji} \chi_{j,t} \quad \dots(8)$$

where L is the lag operator, $\psi_i(L) = 1 - \sum_{j=1}^n \psi_{ij} L^j$ and $\Theta_i(L) = 1 - \sum_{j=1}^s \theta_{ij} L^j$

$D_t = \text{diag}(h_{iit}^{1/2} \dots h_{nnt}^{1/2})'$ is a diagonal matrix of square root conditional variances, where h_{iit} can be defined as any univariate GARCH type models and R_t is the $t * (n(n - 1)/2)$ matrix containing the time varying conditional correlations defined as:

$$R_t = \text{diag}(q_{11,t}^{-1/2} \dots q_{nn,t}^{-1/2}) Q_t \text{diag}(q_{11,t}^{-1/2} \dots q_{nn,t}^{-1/2}) \dots (9)$$

where $Q_t = (q_{ij,t})$ is a $n \times n$ symmetric positive definite matrix given by:

$$Q_t = (1 - \alpha - \beta) \bar{Q} + \alpha u_{t-1} u_{t-1}' + \beta Q_{t-1} \dots (10)$$

where $u_t = (u_{1t}, u_{2t}, \dots, u_{nt})'$ is the $n \times 1$ vector of standardized residuals, $\bar{Q}$ is the $n \times n$ unconditional variance of u_t and α and β are non-negative scalar parameters satisfying $\alpha + \beta < 1$.

All the flexible versions of the MGARCH models are estimated under a multivariate Student t distribution (Harvey, Ruiz and Shephard 1992 and Fiorentini, Sentana and Calzolari 2003) as the normality assumption is rejected in most empirical applications dealing with daily exchange rate data. This view is also endorsed by Bollerslev (1986), Heish (1989) and Baillie and Bollerslev (1989) who find evidence that GARCH (1,1) model with Students t distribution, rather than normal distribution, is the most appropriate model for analyzing exchange rate data.

In order to study the spillover among the returns, an index is calculated on the basis of vector autoregressive (VAR) models popularize by Sims (1980). However, we follow the broad literature of Engle, Ito and Lin (1990). To amalgamate spillover effects across different foreign exchange markets and extract plethora of information into a single spillover measure we have used the technique of variance decomposition. To construct the spillover index, for each market i we simply add the shares of its forecast error variance coming from shocks to asset j , for all $i \neq j$, and then we add across all $i = 1, 2, \dots, N$.

To avoid complexities, we are illustrating the procedure using a two variable covariance stationary VAR process:

$$x_t = \Phi x_{t-1} + \varepsilon_t \quad (11)$$

where $x_t = (x_{1,t}, x_{2,t})'$ and Φ is a 2×2 parameter matrix. Being a covariance stationary VAR process, it can alternatively be presented as a vector moving average process.

$$x_t = \Theta(L) \varepsilon_t \quad (12)$$

where $\Theta(L) = (I - \Phi L)^{-1}$. Equation (2) can alternatively be written as:

$$x_t = A(L) \varepsilon_t \quad (13)$$

where $A(L) = \Theta(L)Q_t^{-1}$, $u_t = Q_t \varepsilon_t$, $E(u_t u_t') = I$, and Q_t^{-1} is the unique lower-triangular Cholesky factor of the covariance matrix of ε_t .

Now, the optimal one step ahead forecast becomes

$$x_{t+1,t} = \Phi x_t \quad (14)$$

and the corresponding one step ahead error vector becomes

$$e_{t+1,t} = x_{t+1} - x_{t+1,t} = A_0 u_{t+1} = \begin{bmatrix} a_{0,11} & a_{0,12} \\ a_{0,21} & a_{0,22} \end{bmatrix} \begin{bmatrix} u_{1,t+1} \\ u_{2,t+1} \end{bmatrix}$$

which has covariance matrix

$$E(e_{t+1,t} e_{t+1,t}') = A_0 A_0' \quad (15)$$

Therefore, the variance of one step ahead error in forecasting of $x_{1,t}$ and $x_{2,t}$ are $a_{0,11}^2 + a_{0,12}^2$, and $a_{0,21}^2 + a_{0,22}^2$, respectively. Variance decomposition helps us to fractionate one step ahead error variance in forecasting x_i due to shocks to x_i and due to x_j , $i \neq j$. Now, $a_{0,21}^2$ and $a_{0,12}^2$ are $x_{1,t}$ shocks that affect the forecast error variance of $x_{2,t}$; and $x_{2,t}$ shocks that affect the forecast error variance of $x_{1,t}$, respectively. Then, total spillover is $a_{0,12}^2 + a_{0,21}^2$. Total spillover index (S) is then calculated by dividing total spillover by total error forecast error variance.

$$S = \frac{a_{0,12}^2 + a_{0,21}^2}{\text{trace}(A_0 A_0')} \cdot 100 \quad (16)$$

where $\text{trace}(A_0 A_0') = a_{0,11}^2 + a_{0,12}^2 + a_{0,21}^2 + a_{0,22}^2$. Now, we can generalize the index by considering p^{th} order N variable VAR using H step ahead forecasts.

$$S = \frac{\sum_{h=0}^{H-1} \sum_{i,j=1}^N a_{h,i,j}^2}{\text{trace}(A_0 A_0')} \cdot 100 \quad (17)$$

3.3 Empirical Results

3.3.1 Descriptive Statistics

Table 1 reports the descriptive statistics of Indian Rupee, Chinese Yuan, Brazilian Real, South African Rand, Euro, Japanese Yen, Australian Dollar and Swiss Franc for the period 1995-2015. The standard deviation values show that the returns of the foreign exchange markets are positive and highest for South Africa followed by Brazil, India and China. This result signifies that variation in South African Rand is more as compared to others emerging economies. In addition, the Jarque-Bera statistics of all exchange rate returns indicates the presence non-normal distribution since all the coefficients are significant at one percent level thus rejecting the null hypothesis of normally distributed returns.

Table 1: Descriptive Statistics

	Indian Rupee	Chinese Yuan	Brazilian Real	South African Rand	Euro	Japanese Yen	Australian Dollar	Swiss Franc
Mean	0.000132	-5.91E-05	0.000247	0.000234	3.90E-05	5.35E-05	-2.99E-05	-7.03E-06
Median	-3.60E-05	-7.61E-06	0.000158	0.000119	8.44E-05	0.000152	0.000137	-0.000180
Maximum	0.038169	0.007256	0.114657	0.083730	0.029926	0.033964	0.045401	0.081317
Minimum	-0.037195	-0.017483	-0.089720	-0.091201	-0.046058	-0.056625	-0.129872	-0.066806
Std. Dev.	0.004318	0.000730	0.009550	0.009724	0.005906	0.006791	0.006869	0.007635
Skewness	0.100569	-3.074689	0.488242	0.329507	-0.170353	-0.472464	-1.417929	0.702707
Kurtosis	14.64613	84.09503	20.12225	9.982374	5.423840	8.047721	30.25439	15.09329
Jarque Bera (Prob.)	29486.23 (0.000)*	1437490 (0.000)*	63923.16 (0.000)*	10690.16 (0.000)*	1302.062 (0.000)*	5731.596 (0.000)*	163183.3 (0.000)*	32213.79 (0.000)*
Q(1)	1.4581 (0.227)	1.2600 (0.262)	4.5274 (0.033)**	2.8137 (0.093)	2.8732 (0.090)	0.7909 (0.374)	0.0090 (0.924)	6.7924 (0.009)*
Q ² (1)	424.31 (0.000)*	37.533 (0.000)*	628.30 (0.000)*	576.10 (0.000)	12.599 (0.000)*	187.01 (0.000)*	178.54 (0.000)*	111.59 (0.000)*
Obs.	5216	5216	5216	5216	5216	5216	5216	5216

Notes: *, ** indicates significance at 1% and 5% level. P-values are in the parentheses.

Q(1) and Q²(1) is the Ljung-Box statistic for serial correlation in return series and squared series respectively

Accordingly the table also reports the Ljung-Box Q and the Q² statistics for all the return series and the squared return series. The Q statistic points to the fact that all the exchange rates can be characterized as random walk processes except the Brazilian Real and Swiss Franc.

Alternatively the Q^2 statistic is significant for each return series indicating the presence of higher order serial correlation and non-linearity among the variables. These findings also strengthen the fact that exchange rate volatility can be modelled in a GARCH framework.

Furthermore the returns series (Figure 2, Appendix) of all the currencies exhibit randomness and volatility clustering which means that large movements are characterized by large changes and vice-versa. This conclusion also finds evidence from the literature that exchange rate volatility can be modelled in a GARCH process.

3.3.2 RETURN CO-MOVEMENT:

In this section we look into the return co-movement in the emerging foreign exchange markets with world's four most traded currencies by applying Dynamic Conditional Correlation model. But before continuing with the estimation, the correlation among the variables is presented in table 2. These correlation coefficients determine the degree to which two variable's movements are associated. The table discloses that Indian Rupee, Chinese Yuan and South African Rand are positively correlated with Euro, Japanese Yen, Australian Dollar and Swiss Franc. On the other hand the Brazilian Real is also positively correlated with Euro, Australian Dollar and Swiss Franc but negatively correlated with Japanese Yen. It is important to note here that almost all the emerging markets have the highest significant positive correlation with Australian Dollar, Euro and Swiss Franc and nearly insignificant correlation with Japanese Yen.

TABLE 2: Cross Correlation

	Euro	Japanese Yen	Australian Dollar	Swiss Franc
Indian Rupee	0.187661 (0.000)*	0.003984 (0.7736)	0.265589 (0.000)*	0.114220 (0.000)*
Chinese Yuan	0.067685 (0.000)*	0.006305 (0.6489)	0.089584 (0.000)*	0.042771 (0.0020)*
Brazilian Real	0.200760 (0.000)*	-0.016091 (0.2453)	0.365206 (0.000)*	0.086949 (0.000)*
South African Rand	0.387282 (0.000)*	0.007100 (0.6082)	0.507310 (0.000)*	0.262830 (0.000)*

Notes: * indicates significance at 1% level. P-values are in the parentheses

The positive correlation symbolizes that an appreciation of one currency leads to an appreciation of other and vice-versa and the negative correlation exhibits that an appreciation of one currency leads to depreciation of other and vice-versa.

We shall now focus on simple AR(1)-MGARCH-DCC(1,1) model for illustrating the return co-movement in the foreign exchange market. Four DCC model was estimated separately to understand the co-movements between the emerging markets with the developed foreign exchange markets, for example DCC was conducted for Indian Rupee, Chinese Yuan, Brazilian Real and South African with the developed country's currency independently. Table 3 depicts the estimates of DCC parameters α and β to be statistically significant indicating that the second moments of exchange rate returns are time varying. Furthermore the summation of the parameters are less than one in all the cases which signifies that the DCC model is very well specified as the stability condition is met.

Table 3: DCC Estimate

Coefficient	DCC Estimate			
	India Rupee	China Yuan	Brazilian Real	South African Rand
α	0.023353*	0.022847*	0.024284*	0.024712*
β	0.974251*	0.974935*	0.973667*	0.973085*
$\alpha + \beta$	0.997604	0.997782	0.997951	0.997797
Stability Condition $\alpha + \beta < 1$	Met	Met	Met	Met

* denotes significance at 1% level

The univariate GARCH estimate of emerging market economies and the developed economies are presented in table 4 and table 5. In the case of emerging market economies, the summation of ARCH and GARCH coefficients are all greater than one which relates that the return series of the emerging foreign exchange markets are explosive thus pointing to the tendency to move away from the mean value whereas the summation of ARCH and GARCH coefficients of the developed economies (Table 5) are less than one which means the volatility of the return series of the developed economies have a leaning to persist for a long time. Furthermore the conditional variances graphs (Figure 3, Appendix) of the respective markets also strengthen the above argument. Nevertheless the ARCH-LM test confirms that no further ARCH effect exists in the returns series except in South African Rand and Euro.

Table 4: Univariate GARCH Estimate and Diagnostic Test: Emerging Market Economies

Countries	ARCH (γ)	GARCH (δ)	$\sum (\gamma + \delta)$	Status of the series	ARCH-LM test
Indian Rupee	0.254493*	0.808864*	1.063357	Explosive	0.003253 (0.8143)
Chinese Yuan	0.431677*	0.738855*	1.170532	Explosive	-0.000198 (0.9886)
Brazilian Real	0.164723*	0.869247*	1.03397	Explosive	-0.000356 (0.9795)
South African Rand	0.104427*	0.909714*	1.014141	Explosive	0.037116 (0.0073)*

* denotes significance at 1% level

Table 5: Univariate GARCH Estimate and Diagnostic Test: Developed Economies

Countries	ARCH (γ)	GARCH (δ)	$\sum (\gamma + \delta)$	Status of the series	ARCH-LM test
Euro	0.031710*	0.967370*	0.99908	Persist for long time	-0.030392 (0.0282)**
Japanese Yen	0.037300*	0.954671*	0.991971	Persist for long time	0.024158 (0.0811)
Australian Dollar	0.041354*	0.953699*	0.995053	Persist for long time	-0.002195 (0.8741)
Swiss Franc	0.029869*	0.964356*	0.994225	Persist for long time	0.004962 (0.7202)

* and ** denotes significance at 1% and 5% level

Besides this the conditional correlation graphs (Figure 4, Appendix) report evidence of significant dynamic conditional correlation among the emerging markets and developed markets. It is indicative that there is high positive correlation between Indian Rupee, Brazilian Real and South African Rand with Euro and Australian Dollar, but the degree of correlation, though positive, is small in case of Chinese Yuan. This high correlation with Euro and Australian Dollar is particularly during the global recession in 2008. The correlation further signifies that Indian Rupee and Chinese Yuan have a considerable co-movement with Euro while Brazilian Real and South African Rand have a considerable co-movement with Australian Dollar. It can also be argued that the implied volatility of Euro and Australian Dollar significantly affects the volatility expectations of Rupee, Yuan, Real and Rand.

3.3.3 VOLATILITY SPILLOVER:

In this section, the estimate of the volatility spillover is presented based on vector autoregression, variance decomposition framework and spillover index of Diebold and Yilmaz (2009). The results of the degree and direction of volatility spillovers within and across the four emerging market economies are shown in table 6, table 7, table 8 and table 9 respectively. Several interesting results emerge in the tables. In all the cases, the own market volatility spillover is highest with the diagonal elements having higher values as compared to off-diagonal elements. In case of Indian Rupee and Chinese Yuan, the highest spillover is from Euro followed by Australian Dollar whereas in case of Brazilian Real and South African Rand, the spillover is highest from Australian Dollar followed by Euro. So it can be inferred that Euro is the dominant currency in volatility transmission to the Asian emerging market economies whereas Australian Dollar is the dominant currency in volatility transmission of Brazilian Real and South African Rand. The estimates of the table below further strengthen the above discussion.

Table 6: Indian Rupee

	From					
	Indian Rupee	Euro	Japanese Yen	Australian Dollar	Swiss Franc	Contribution From Others
To						
Indian Rupee	99.54847	0.32873	0.03419	0.08397	0.00463	0.45153
Euro	3.58035	96.07010	0.00918	0.09362	0.24675	3.92990
Japanese Yen	0.03549	0.00392	99.80584	0.12277	0.03200	0.19416
Australian Dollar	7.48457	20.23964	0.06704	72.07212	0.13663	27.92788
Swiss Franc	1.35285	63.73844	0.08756	0.42047	34.40069	65.59931
Contribution to Others	12.45325	84.31073	0.19797	0.72082	0.42001	98.10278
Contribution including own	112.00172	180.38083	100.00381	72.79294	34.82070	
Net Spillover	12.00173	80.38083	0.00381	-27.20706	-65.17931	Spillover Index =19.620556%

Table 7: Chinese Yuan

	From					
	Chinese Yuan	Euro	Japanese Yen	Australian Dollar	Swiss Franc	Contribution From Others
To						
Chinese Yuan	97.93280	1.48771	0.17822	0.19871	0.20256	2.06720
Euro	0.50769	98.92615	0.01053	0.13044	0.42520	1.07386
Japanese Yen	0.02226	0.16528	99.40034	0.32847	0.08365	0.59966
Australian Dollar	0.87056	23.96154	0.08751	74.77777	0.30262	25.22223
Swiss Franc	0.32167	64.60138	0.08496	0.58490	34.40709	65.59291
Contribution to Others	1.72219	90.21592	0.36121	1.24251	1.01403	94.55585
Contribution including own	99.65499	189.14207	99.76155	76.02028	35.42112	
Net Spillover	-0.34501	89.14206	-0.23845	-23.97972	-64.57888	Spillover Index =18.91117%

Table 8: Brazilian Real

	From					
	Brazilian Real	Euro	Japanese Yen	Australian Dollar	Swiss Franc	Contribution From Others
To						
Brazilian Real	99.64747	0.04576	0.10520	0.16844	0.03314	0.35253
Euro	4.07923	95.54212	0.00829	0.10546	0.02649	4.21947
Japanese Yen	0.04516	0.00042	99.82119	0.10224	0.03100	0.17881
Australian Dollar	13.69500	18.23807	0.04856	67.85149	0.16688	32.14850
Swiss Franc	0.83774	64.73884	0.08500	0.34719	33.99123	66.00877
Contribution to Others	18.65712	83.02308	0.24705	0.72333	0.25750	102.90808
Contribution including own	118.30459	178.56520	100.06824	68.57482	34.24873	
Net Spillover	18.30460	78.80361	0.06824	-31.42518	-66.26626	Spillover Index =20.58162%

Table 9: South African Rand

To	From					
	South African Rand	Euro	Japanese Yen	Australian Dollar	Swiss Franc	Contribution From Others
South African Rand	99.42328	0.07162	0.04914	0.43168	0.02429	0.57672
Euro	15.00445	84.57841	0.00848	0.09260	0.31606	15.42159
Japanese Yen	0.01788	0.05453	99.76204	0.13056	0.03498	0.23796
Australian Dollar	25.83113	10.28935	0.09791	63.62149	0.16011	36.37851
Swiss Franc	6.96392	58.35267	0.08655	0.26720	34.32965	65.67034
Contribution to Others	47.81738	68.76816	0.24209	0.92204	0.53544	118.28511
Contribution including own	147.24066	153.34657	100.00413	64.54353	34.86509	
Net Spillover	47.24066	53.34658	0.00413	-35.45646	-65.13491	Spillover Index =23.657022%

Furthermore it is also revealed from the corresponding table that the spillover index is highest for South African Rand followed by Brazilian Real, Indian Rupee and Chinese Yuan. In fact if the whole table is considered then a look at the ‘contribution to others’ row reflects that the gross directional volatility spillover to others is highest for Euro and lowest for Japanese Yen in all the cases. However a look at the net volatility spillover row reflects the fact that Euro is the dominant currency in volatility transmission and Swiss Franc is the net receiver of volatility among other markets. Moreover among the emerging markets, it is observed that only Chinese Yuan is the net receiver of volatility while Indian Rupee, Brazilian Real and South African Rand are the net transmitter of volatility to the developed economies.

4. CONCLUSION AND POLICY IMPLICATION

This paper has investigated the return co-movement and volatility spillover among the emerging market economies and the developed economies. To recapitulate, exchange rate is one of the critical variables that occupy a central position in policy proposals and whose stability is considered as one of the most important parameters in achieving high economic growth. Ever since the exchange rate became market determined, the pattern of volatility has undergone a distinct change and the possibility of such volatility transmission had multiplied. The period

from 2000 onwards witnessed sustained volatility in the exchange rate of developing economies. This necessitated a detailed econometric analysis of the return co-movement and volatility spillover.

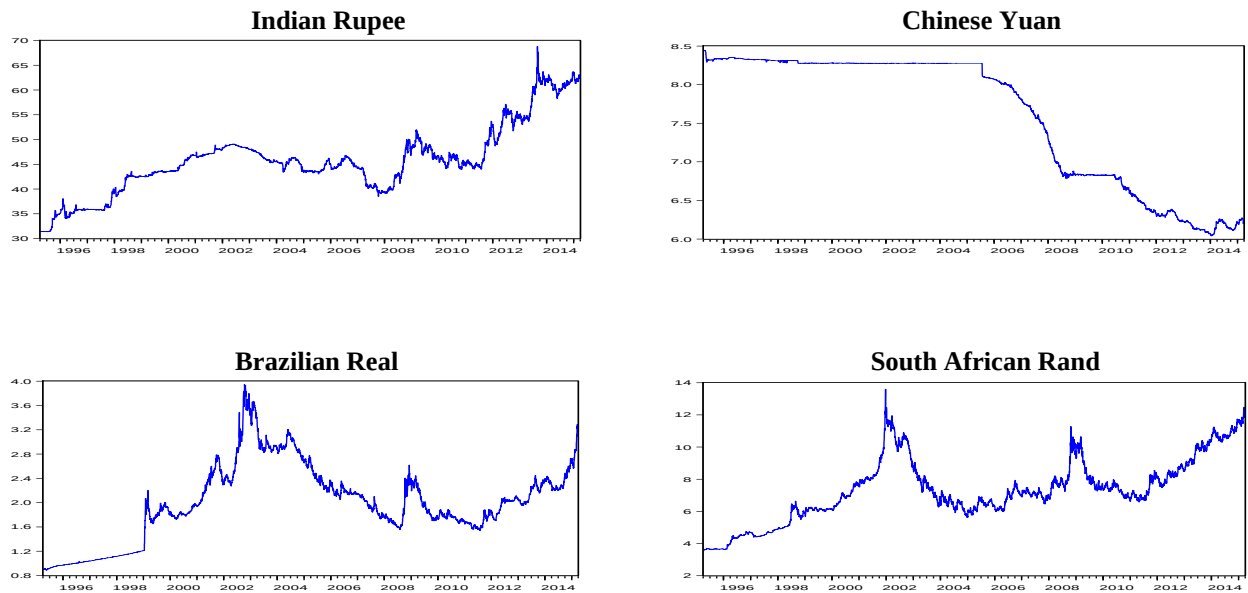
The econometric analysis had set out in identifying that most of the emerging foreign exchange markets are explosive compared to the developed economies. It is also inferred from the analysis that Indian Rupee and Chinese Yuan are more influenced by Euro whereas Brazilian Real and South African Rand are influenced more by Australian Dollar among others. Apart from this it is also revealed from the table that Euro is the net transmitter of volatility to Indian Rupee and Chinese Yuan and Australian Dollar is the net transmitter of volatility to Brazilian Real and South African Rand. However, dynamic correlations and volatility spillovers show large variability and are positively associated with extreme economic episodes, such as during the global recession.

APPENDIX

LIST OF FIGURES

FIGURE 1: TIME PATH OF THE VARIABLES

EMERGING MARKET ECONOMICS



DEVELOPED ECONOMIES

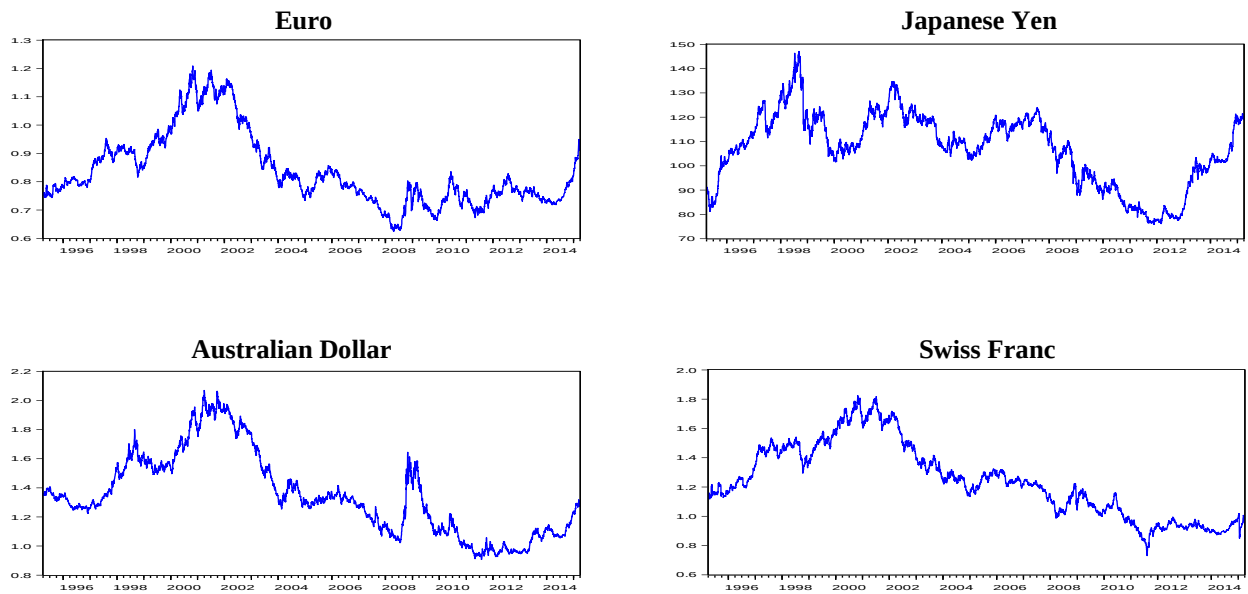
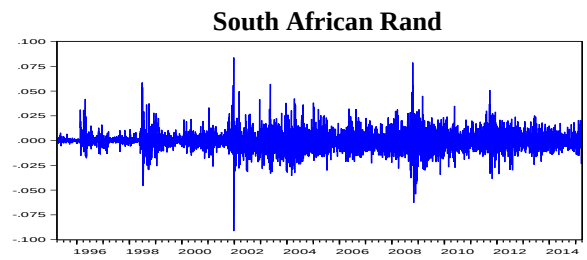
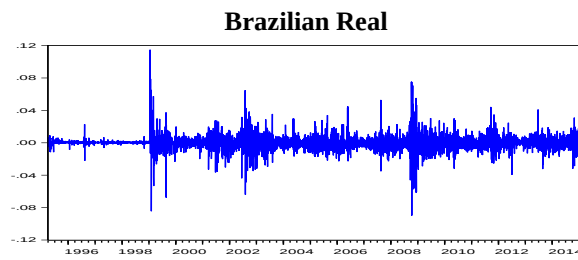
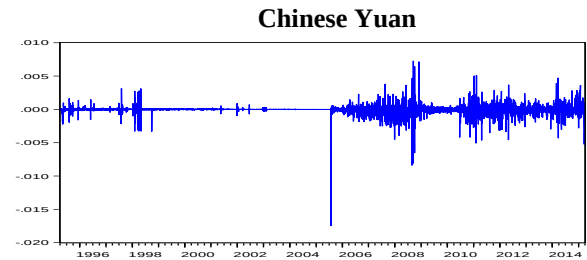
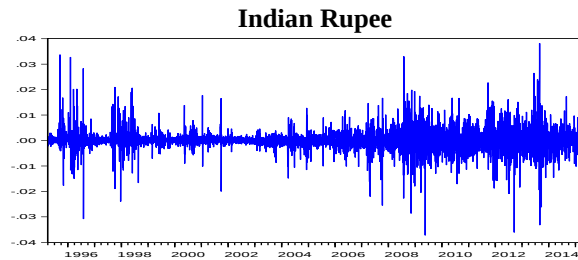


FIGURE 2: RETURN SERIES

EMERGING MARKET ECONOMIES



DEVELOPED ECONOMIES

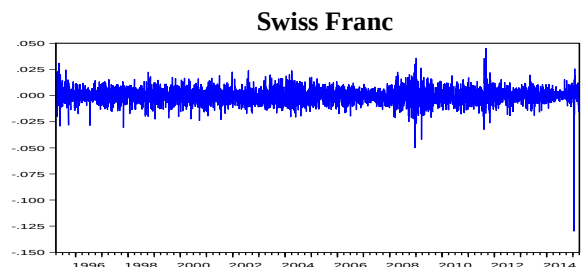
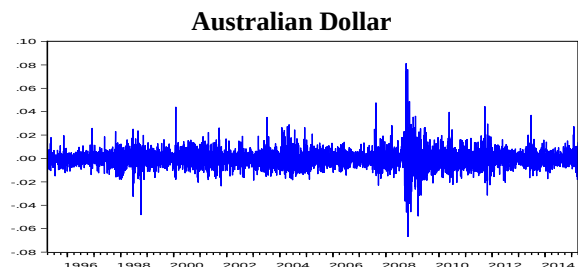
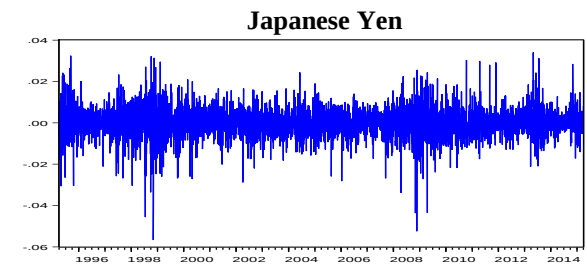
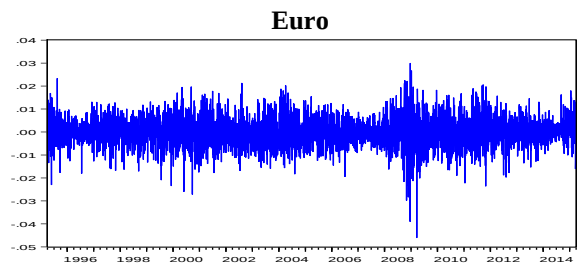
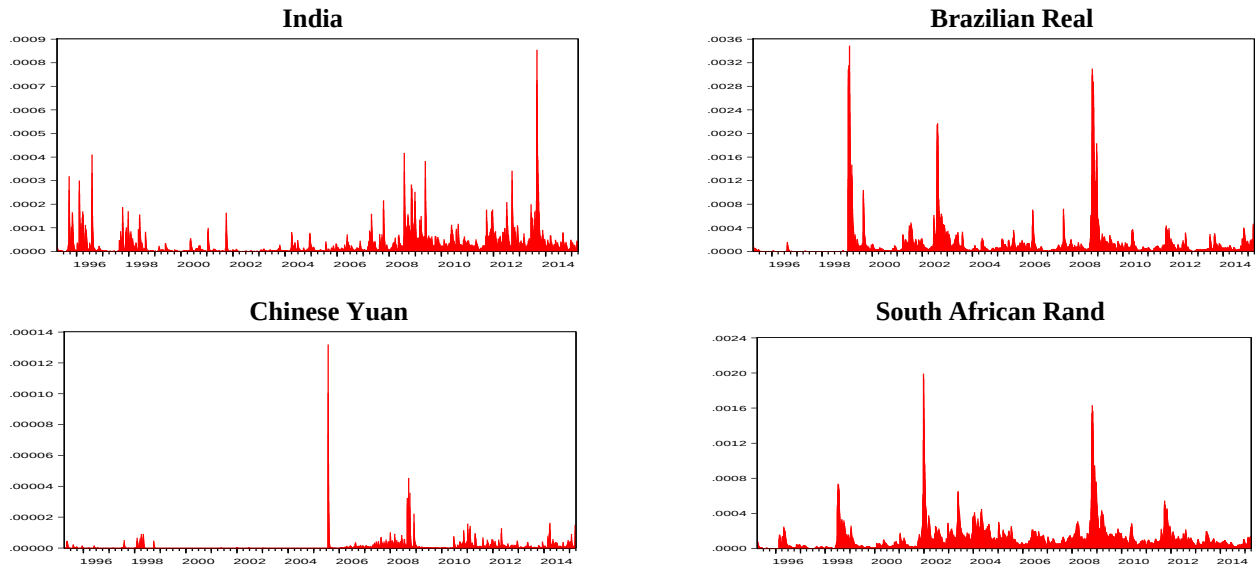


FIGURE 3

**CONDITIONAL VARIANCE
EMERGING MARKET ECONOMIES**



DEVELOPED ECONOMIES

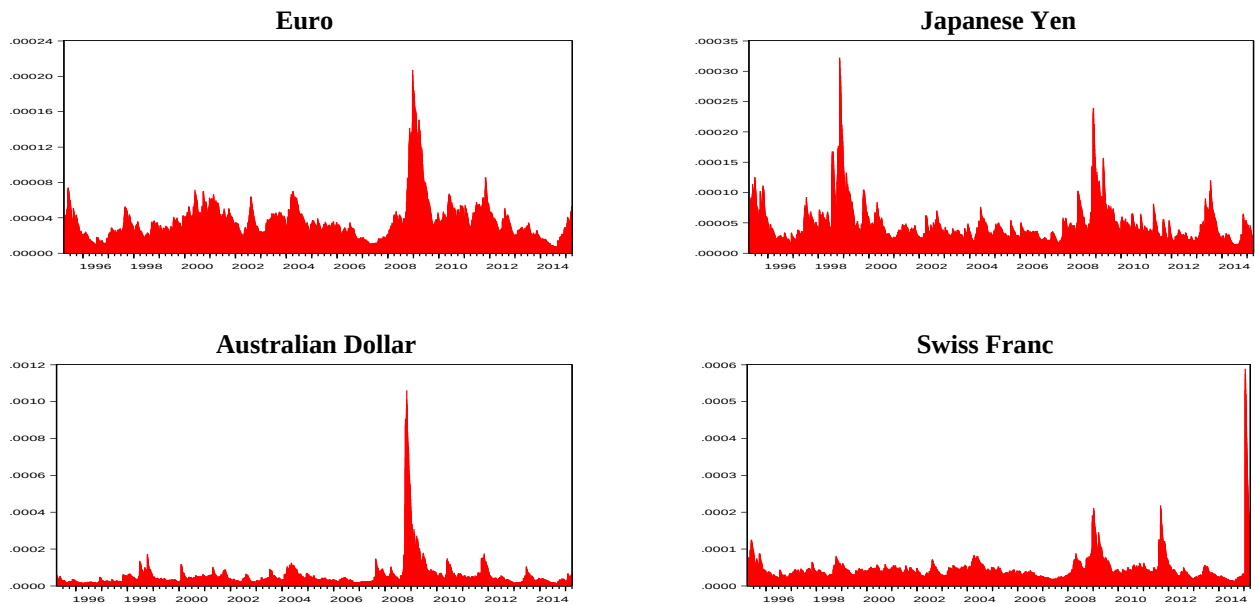
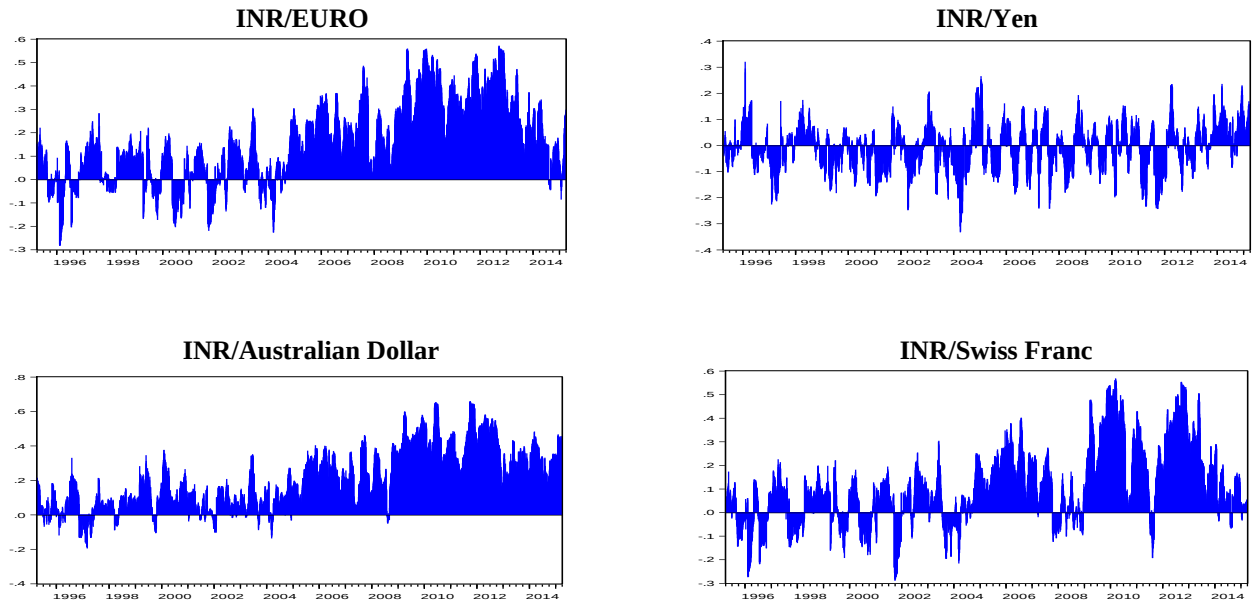
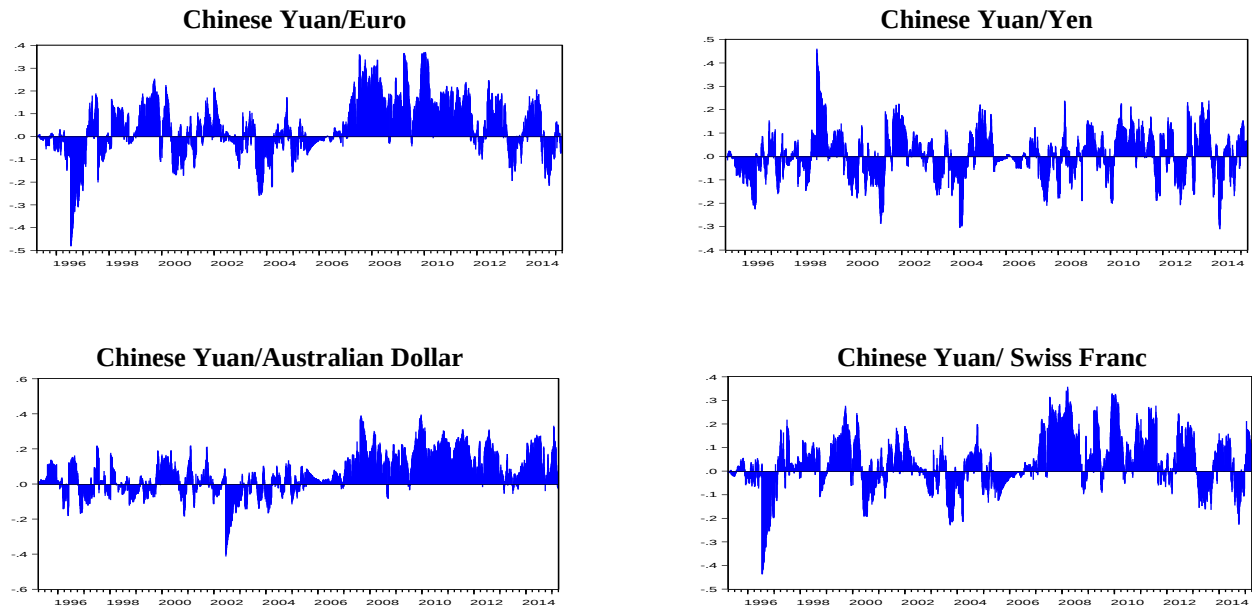


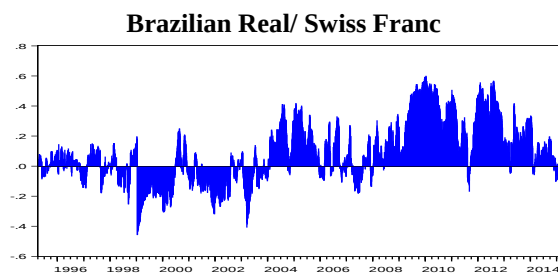
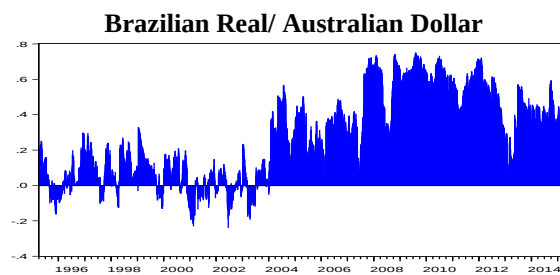
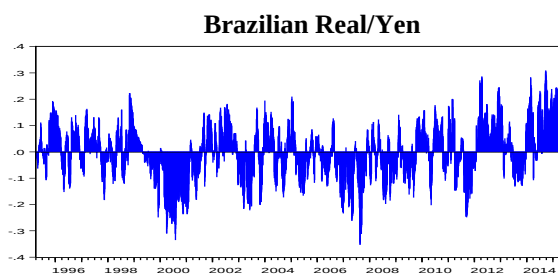
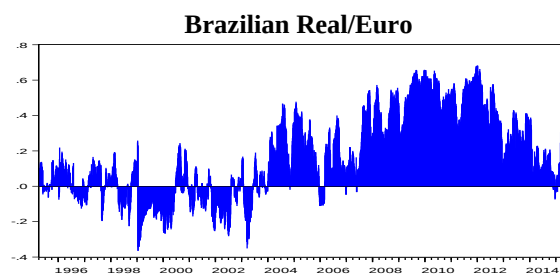
FIGURE 4
CORRELATION
INDIAN RUPEE



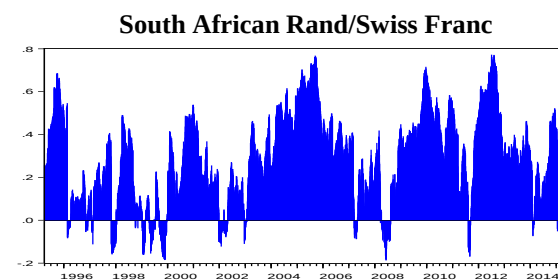
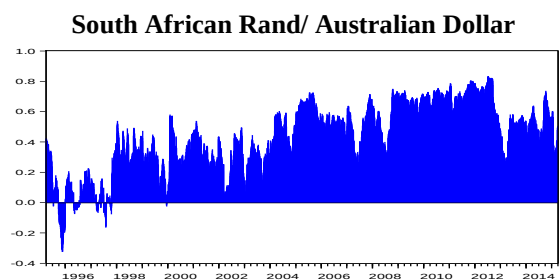
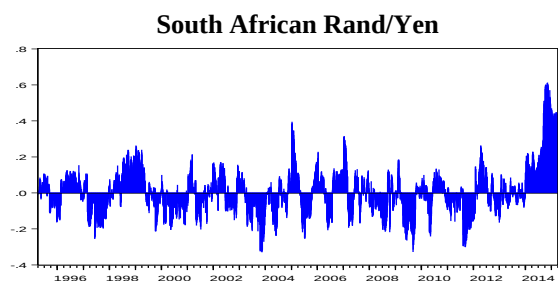
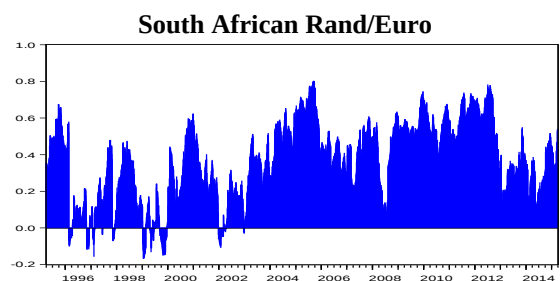
CHINESE YUAN



BRAZILIAN REAL



SOUTH AFRICAN RAND



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